

Concordia University

Personal Task Management System

Iteration 1

SOEN 342: Winter 2026

System Operations

And

System Contracts

System Operations:

Operation Signature
CreateTask(creation_date, priority, status, title, description, due_date)
createProject(name, description)
addSubtask(taskId, title)
completeSubtask(taskId, subtaskId)
UpdateTask(title, attribute, newValue)
assignTaskToProject(taskId, projectId)
moveTaskToProject(taskId, newProjectId)
removeTaskFromProject(taskId)
addTag(title, keyword)
removeTagFromTask(taskId, tag)
viewTask()

Operation Contracts:

contract CO1: CreateTask

Operation:	CreateTask(creation_date: Date, priority: PriorityLevel, status: Status, title: String, description: String, due_date: Date)
Cross Ref:	Use Case: Task Creation
Preconditions:	1. title is not null and not empty. 2. priority is a valid PriorityLevel value. 3. No task with the same title already exists in the system.
Postconditions:	- An instance of task, t, was created. - An instance of history, h, was created.

	- t.title was set to title. - If description was provided, t.description was set to description. [Conditional] - t.priority_level was set to priority. - If due_date was provided, t.due_date was set to due_date. [Conditional] - t.creation_date was set to the current system date/time. - t.status was set to open. - h.timestamp was set to the current system date/time. - h.description was set to "Task created". - t was associated with the system's task collection. - h was associated with t.
--	---

Contract CO2: createProject

Operation:	createProject(name: String, description: String)
Cross Ref:	Use Case: Project Creation
Preconditions:	1. name is not null and not empty.
Postconditions:	- An instance of project, p, was created. - p.name was set to name. - If description was provided, p.description was set to description. [Conditional] - p was associated with the system's project collection.

Contract CO3: addSubtask

Operation:	addSubtask(taskId: ID, title: String)
Cross Ref:	Use Case: Subtask Creation
Preconditions:	1. A task instance t with t.id = taskId exists in the system. 2. title is not null and not empty.

Postconditions:	- An instance of subtask, s, was created. - s.title was set to title. - s.completion_status was set to incomplete. - s was associated with task t. - s was added to t's collection of subtasks.
------------------------	---

Contract CO4: completeSubtask

Operation:	completeSubtask(taskId: ID, subtaskId: ID)
Cross Ref:	Use Case: Subtask Creation
Preconditions:	1. A task instance t with t.id = taskId exists in the system. 2. A subtask instance s with s.id = subtaskId exists and is associated with t. 3. s.completion_status = incomplete.
Postconditions:	- s.completion_status was set to complete.

Contract CO5: UpdateTask

Operation:	UpdateTask(title: String, attribute: FieldName, newValue: Object)
Cross Ref:	Use Case: Update Tasks
Preconditions:	1. A task instance t with t.title = title exists in the system. 2. attribute is a valid field name. 3. newValue is valid for the given attribute. 4. If attribute = project: a project with the given value exists in the system. 5. If attribute = tags: a tag with the given value exists in the system.
Postconditions:	- An instance of history, h, was created. - The attribute identified by attribute on t was set to newValue. - h.timestamp was set to the current system date/time.

	- h.description was set to a summary of the change. - h was associated with t.
--	---

Contract CO6: assignTaskToProject

Operation:	assignTaskToProject(taskId: ID, projectId: ID)
Cross Ref:	Use Case: Update Tasks
Preconditions:	1. A task instance t with t.id = taskId exists in the system. 2. A project instance p with p.id = projectId exists in the system. 3. t is not currently associated with any project. 4. t is not a subtask instance.
Postconditions:	- t was associated with p. - t was added to p's collection of tasks.

Contract CO7: moveTaskToProject

Operation:	moveTaskToProject(taskId: ID, newProjectId: ID)
Cross Ref:	Use Case: Update Tasks
Preconditions:	1. A task instance t with t.id = taskId exists in the system. 2. t is currently associated with a project. 3. A project instance p with p.id = newProjectId exists in the system. 4. t is not already associated with p.
Postconditions:	- t was dissociated from its previous project. - t was associated with p. - t was added to p's collection of tasks.

Contract CO8: removeTaskFromProject

Operation:	removeTaskFromProject(taskId: ID)
Cross-Ref:	Use Case: Update Tasks
Preconditions:	1. A task instance t with t.id = taskId exists in the system. 2. t is currently associated with a project.
Postconditions:	- t.project was set to null. - t was removed from the project's collection of tasks.

Contract CO9: addTag

Operation:	addTag(title: String, keyword: String)
Cross Ref:	Use Case: Task Tags
Preconditions:	1. A task instance t with t.title = title exists in the system. 2. keyword is not null and not empty. 3. No tags instance with name = keyword is already associated with t.
Postconditions:	- An instance of tags, tg, was created with tg.name = keyword, if no such instance existed. - tg was associated with t.

Contract CO10: removeTagFromTask

Operation:	removeTagFromTask(taskId: ID, tag: String)
Cross Ref:	Use Case: Task Tags
Preconditions:	1. A task instance t with t.id = taskId exists in the system. 2. A tags instance tg with tg.name = tag is associated with t.
Postconditions:	- tg was dissociated from t.